LEF MODEL · CHAPTER 8

# Chapter 8: What Is a Mare Really? — the Feeding Openings, the 73 Gravity Domes and the Floor That Sank Like a Frozen Lake

Based on Paper 05 — The Maria (Parts A–F: Studies 1–7 and 16)

01 · THE MYSTERY

## The mystery: why does the Moon have two faces — and where are the maria of the far side hiding?

Anyone who looks at the Moon, even through simple binoculars, at once notices the sharp contrast that shaped the "man in the Moon": **the near side**, always facing the Earth, is covered with dark, round expanses ("maria" — *Maria* in Latin, because the ancients thought they were oceans), while **the far side** is bright, scarred and almost free of dark patches.

The accepted literature interprets each mare as two separate, unrelated events: first a giant asteroid struck the Moon and dug a deep basin in it; and hundreds of millions of years later dark basaltic lava broke out of the mantle and flooded the basin. By this definition, a mare is simply **a surface of dark basalt**.

But when the precise gravity maps of the GRAIL spacecraft are examined, a mystery appears that shakes this story to its foundations: beneath the bright surface of the far side hide **dozens of round, dense bodies of excess gravity** — identical in size and strength to those beneath the giant maria of the near side, but without any dark basalt on their surface! What are these hidden bodies, and does the far side look mare-free only because we are looking at its colour?

02 · AN IMAGE

## An image: covered wells in a field and a frozen crust on a lava lake

To understand the turnaround that Paper 05 proposes in the definition of the maria, imagine two simple images:

- **A lava lake in a volcano's crater:** looking down on an extinct volcanic crater, you see a dark, frozen rock surface. This dark crust is not the mountain itself — it is only the last layer that froze from above after the flow stopped. Beneath it lies the deep vent through which the lava rose. In exactly the same way, **the dark basalt on the Moon's face is not the mare — the mare is the opening through which the Moon was built**, and the basalt is only the last crust that froze at the top of the opening!
- **A field with active wells on two sides — when one side has been covered with earth:** imagine a large field in which dozens of wells were dug. On one side of the field (the near side), the wells stayed open until the end of the building, so the dark water surfaces inside them can be seen. On the other side of the field (the far side), the wells closed earlier and were paved over with a thick layer of bright earth. Someone who counts only "dark water patches" will think there are wells on one side only — but an instrument that measures below the ground will find that the covered side has even **more wells**!

03 · SCIENCE AND MODEL

## Science and model: separating the concepts and counting the 73 feeding domes

Parts A–D of Paper 05 break the word "mare" into four clear physical components, from the bottom up:

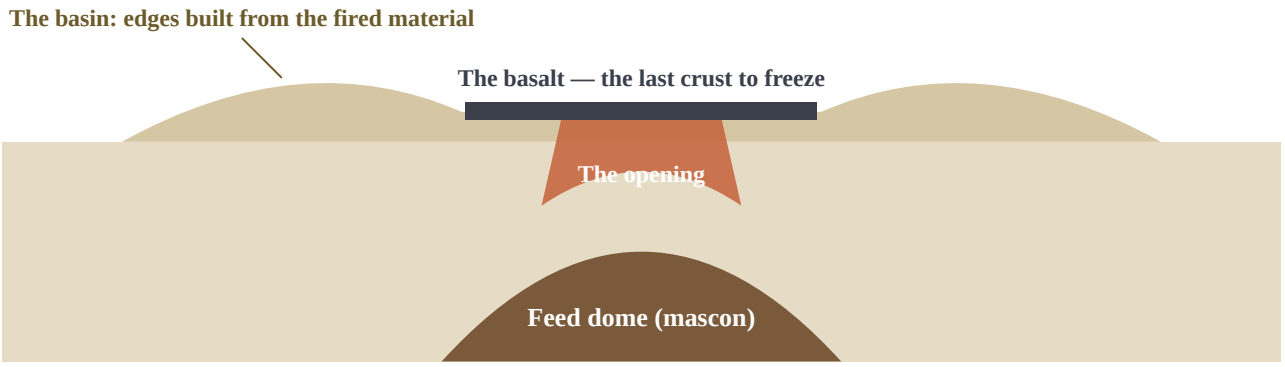
1. **The feed dome:** a dense body rich in iron and titanium that was pushed upward from an underground focus, lifted its centre and widened its edges. When the feeding ended, the dome froze into a dense body that pulls the

GRAIL spacecraft more strongly — and this is the **mascon** (mass concentration) measured in gravity.

2. **The opening:** the fluid, low-viscosity zone in the shell through which material rose to the surface. Through most of the openings, material was fired outward and fell back onto the Moon to create the craters (**exit openings**); while through **Mare Imbrium** and **Mare Serenitatis** — which sat inside the contact disc with the Earth — the pressure and material passed inward into the Moon without firing outward (**feeding mouths**).

3. **The mare basalt:** the last layer that froze at the top of the opening. By contrast, broad basalt surfaces with no feed dome beneath them — such as **Oceanus Procellarum**, Mare Tranquillitatis and Mare Fecunditatis — are not "maria" in the structural sense, but **crustal plains melted in place** by the heat of the nearby Earth!

4. **The basin:** the raised rim surrounding the opening — built layer upon layer from the hot material that the opening itself fired around it.




The four components of a mare, from the bottom up: feed dome, opening, basalt and basin. (Schematic drawing.)

The decisive discovery in gravity (Studies 1, 3 and 4)

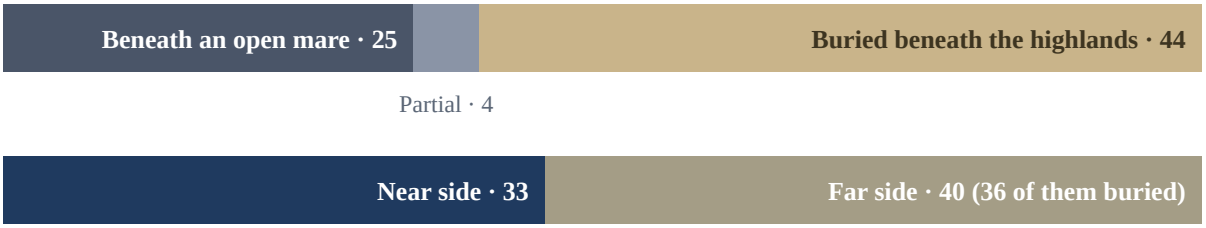
When the GRAIL gravity map (Bouguer anomaly, degrees 6–660) is scanned automatically for every round dense body 100 km or more across and 50 mGal or more in strength, **73 feeding domes** appear on the face of the Moon:

• **33 of the 34 maria** identified on the gravity-gradient map sit exactly on one of the 73 domes (in 20,000 random runs not a single run reached 33!). The only one not sitting on a dome is **Mare Ingenii**, whose dome was "cooked" and erased by the enormous heat beneath the South Pole–Aitken basin.

• Of the 73 domes, 25 lie beneath an open mare, 4 are partly covered, and **44 domes are completely buried beneath the bright highlands — 36 of them on the far side**! That is, the far side has **more feeding domes than the near side (40 against 33)**.

• **The crustal step (17 km on average):** beneath the centre of the opening the crust is 8–29 km thinner (median 17 km) than in the ring around it (in 29 of 36 marked openings, compared with only 2% in ordinary impact craters) — proof that the dome pushed and lifted from below, not an external impact from above.

• **The blanket over the buried domes:** above the 44 buried domes, the iron content (−0.19 wt%) and thorium (+0.05 ppm) are completely identical to the surrounding ground, and the crust is thicker by **8–13 km** — the thickness of the later layers that covered the ancient openings of the far side together with their craters!




73 feeding domes in the GRAIL scan — by cover (top) and by side (bottom)

The far side has more feeding domes than the near side; most of them are buried beneath the bright highlands and so do not look like maria.

04 · THE MECHANISM IN WORDS

## The mechanism in words: the floor of a frozen lake, the sinking of the centre and the collar of fired material

What happens to the floor of a mare once the feeding through it stops, and why is it shaped like a sunken basin surrounded by raised edges? Parts E and F of Paper 05 reveal four measurable geometric phenomena:

1. **One level, like a frozen lake (Study 6):** a fluid at rest under gravity cannot hold a slope — it settles to a single uniform level and freezes that way. In precise height measurements (SLDEM2015), **16 out of 16 large mare floors** (and also 188 of the 189 mapped basalt units) show **a single height level**, lying deep below the rim of the mare (median 3.4 km, chance probability 1 in 65,000) and below the surrounding ground (median 1.6 km).

2. **The centre sank above the dome — except in the two contact maria (Study 5):** when the feed dome freezes into a dense, iron-rich body, the centre above it sinks (mainly through thermal contraction on cooling, alongside the weight of the dense body and suction). Indeed, in **10 of the 12 open maria** the centre is lower than the edges (median −267 m, for example −2.2 km at Orientale and −1.6 km at Nectaris), and the stronger the gravity dome, the deeper the sinking ($\rho = -0.59, p = 0.04$).
   **The only two exceptions, which confirm the rule:** at **Mare Serenitatis (+60 m)** and **Mare Imbrium (+43 m)** the centre is actually slightly higher than the edges! Why? Because these are the two maria that sat inside the contact

disc and froze very close to the Earth, whose strong pull held the near side stretched outward while they froze!

3. **Stretched edges and a compressed interior (§F.2):** like a sheet of ice on a pond whose water drops in the middle, the sinking of the centre stretched the edges of the mare and created rifts in them (**grabens** at 0.76–1.22 radii), and compressed the interior to form **wrinkle ridges** (at 0.71–0.77 radii).

4. **The collar of fired material around the far-side maria (Study 16):** why do the edges around the far-side maria rise so high? Not because lava spilled over the rim (basalt covers only 0%–7% of the ring around them), but because the hot material fired from the opening landed around it and was smeared on layer upon layer! As a result, a raised **"collar"** was built around the far-side maria at 150–300 km (with a median height of +742 m around the buried maria), while around the contact maria — which fed inward and did not fire outward — there is no collar at all (−137 m).

05 · THE KEY EQUATIONS

## Academic explanation and the key equation

The relation between the rate of the flux rising through the opening and its closure, and the balance of craters seen on the surface, are described by the following equations:

$$N_{\text{obs}} = N_{\text{bg}} + N_{\text{return}} - N_{\text{erased}}$$

$$M = \int Q \, dt$$

$$Q_{\text{feed}} = Q_{\text{retained}} + Q_{\text{erupted}}$$

$$t_{\text{shutdown}} \geq t_{\text{exp}}$$

The crater equation ($N_{\text{obs}}$) establishes that a crater count is not a "simple clock": every active opening erased and melted the craters that landed on its floor

($N_{\text{erased}}$) while it fired material that created craters around it ($N_{\text{return}}$). Only from the moment the opening's floor froze did it begin to accumulate new craters — from the last material that kept arriving from the Earth ("the last scatterer", §F.6).

06 · THE SIGNATURE

## The signature in reality: clean floors that rank the order of freezing

If a mare floor begins to keep craters only from the moment it froze, then the number of craters on it serves as a direct measure of **the order in which the maria froze** (Studies 7 and 12):

- **Floors 12 times cleaner than their surroundings:** in all 10 open maria measured (after removing the fields of young rays on both sides), the floor carries on average only **8% (1/12)** of the density of 2–5-km craters in the surrounding ground!
- **A ratio of 83 times in large craters (10 km and more):** the contact maria (Imbrium and Serenitatis) froze last and are therefore the cleanest (256 craters of 2–5 km per million km², and only 3.2 craters of 10+ km per million km²). By contrast, the maria on the hill carry 33 times more 10+-km craters (106 per million km²), and the maria beyond it (Orientale, Moscoviense and Poincaré) carry **83 times** more (266 per million km²)!

### The lab box: counting the domes, the gravity tests and the cleanness of the floors (Paper 05, Parts C–F)

| Measure / study | Sample and measurement conditions | Measured result | Significance / meaning |
|---|---|---|---|
| **Matching maria to feeding domes (Study 1)** | **34 maria on gravity gradients against 73 GRAIL domes** | **33 of 34 maria sit on a dome** | **In 20,000 random runs none reached 33** |
| **Distribution of the 73 feeding domes (Study 4)** | Automatic scan (≥ 100 km, ≥ 50 mGal) | **44 buried** (36 on the far side), 25 open, 4 partial | The far side has more domes (40) than the near side (33) |

| | | | |
|---|---|---|---|
| **The crustal step beneath the openings (Study 3)** | 36 marked openings against 169 control craters | A step of ≥ 8 km in **29 of 36** (median 17 km) | In control craters: only 4 of 169 (2%, median 1 km) |
| **Thickness of the blanket over buried domes (Study 4)** | 44 buried domes against 25 open domes | Crust thicker by **8–13 km**; $\Delta\mathrm{FeO} = -0.19$ wt% | The blanket is identical to the surrounding ground and covered the ancient openings |
| **Floor level and sinking of the centre (Studies 5–6)** | **16 large floors; 12 open maria** | **A single level in 16/16 (3.4 km below the rim); sinking in 10/12** | **Correlation $\rho = -0.59$ (p = 0.04); Imbrium and Serenitatis raised** |
| **Cleanness of the floors of craters (Studies 7 and 12)** | 10 open maria (without ray fields) | Median 8% of 2–5-km craters relative to the surroundings; 83 times between contact and far (10+ km) | The contact maria froze last; the Earth was the last scatterer |

07 · NEXT

# The door ahead

We have discovered that the Moon's maria are not lava lakes that happened to break into asteroid craters, but **the ancient feeding openings** through which the Moon was built: 73 gravity domes (44 of them buried beneath the far-side highlands), each of which left a uniform floor like a frozen lake that sank at its centre after the flow stopped.

But what do the craters that landed on the floors and plains of the maria over time look like — and why are the largest of them melted, the medium ones smeared and the small ones sharp? In the next chapter — **Chapter 9: The Clock on the Floor — Melted, Smeared and Sharp Craters, and the Three Generations of Exposure** (from Paper 05 — *The Maria*, Parts G–I) — we will learn to read the clock of temperature and viscosity on the face of the Moon!

Chapter 9

The Clock on the Floor — Melted, Smeared and Sharp Craters, and the Three Generations of Exposure →

← Chapter 7 · Prologue

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